

ISLINGTON COLLEGE

Intelligent Academic Planning

Challenge Brief

CHALLENGE

Automating Academic Scheduling & Resource Allocation

1. The Problem

Islington College's Routine, Timetable and Examination (RTE) Department manages complex timetables, examination schedules, seating arrangements, classrooms, and academic resources across multiple degree programmes, student cohorts, lecturers, and physical venues. Currently, these operational activities involve significant manual coordination through disconnected spreadsheets and physical logs. This reliance on manual scheduling makes the process exceptionally time-consuming and prone to frequent room clashes, double-booked lecturers, student group timing overlaps, capacity overflow issues, and inefficient venue utilization. The objective of this challenge is to design and develop a modern, scalable, web-based platform that automates, optimizes, and centralizes academic scheduling, exam management, seating allocation, and classroom resource planning into a single intelligent operational hub.

Core Question: How might we design and develop a unified web platform that automates academic scheduling, optimizes examination seating, and eliminates resource conflicts across the RTE Department?

2. What the Platform Should Enable

- **Automate Dynamic Timetabling:** Create and visualize academic schedules with real-time detection of room double-bookings, lecturer clashes, and student cohort overlaps.
- **Digitize Examination Planning:** Exam session setup, module mapping, invigilator assignments, and venue scheduling through a constraint-aware management engine.
- **Track Faculty Workloads and Allocations:** Offer visibility into teaching responsibilities, module assignments, weekly contact hours, and potential scheduling conflicts or overloads.
- **Centralize Physical Resource Tracking:** Monitor classroom inventory, maximum capacities, resource availability, live allocations, and historical room utilization rates.
- **Deliver Student and Staff Lookups:** Provide personalized schedule views and room/seat search tools for students, invigilator duty rosters for faculty, and master allocation views for administrators.
- **Leverage Intelligent Scheduling Optimization:** Apply automated constraint-solving algorithms to recommend clash-free schedules, predict bottleneck trends, and maximize venue efficiency.

3. Proposed Platform Architecture & Workflow

Participants may approach the problem in different ways depending on the area they believe provides the greatest operational impact. Possible areas of focus include:

Academic timetable creation and conflict detection

- Examination scheduling and venue allocation
- Faculty workload and teaching allocation
- Classroom and academic resource management
- Automated or constraint-based scheduling
- Scheduling analytics and optimization

Teams are not required to implement all areas. Solutions should demonstrate a clear approach to solving one or more of the identified scheduling and resource allocation challenges.

4. Core Problem Areas

The challenge contains several interconnected academic scheduling problems that teams may choose to address individually or through an integrated solution:

- **Timetable Management:** Managing classes, lecturers, student cohorts, rooms, and time slots while avoiding scheduling conflicts.
- **Examination Planning:** Coordinating exam dates, modules, venues, invigilators, and student groups.
- **Faculty Workload Management:** Providing visibility into teaching assignments, contact hours, workload distribution, and scheduling conflicts.
- **Classroom & Resource Management:** Managing room capacity, availability, bookings, equipment, and utilization.

Teams may prioritize the problem areas that best support their proposed solution.

5. Suggested Features

The following features are examples only and are not mandatory:

- Visual timetable or scheduling interface
- Clash and double-booking detection
- Room availability and capacity checking
- Faculty workload visualization
- Examination schedule management
- Venue or invigilator allocation
- Schedule search and filtering
- Utilization analytics
- Automated scheduling recommendations

Teams are encouraged to determine which features are most valuable for demonstrating their solution rather than attempting to implement every feature listed.

6. Data Relationships

Participants should determine an appropriate data model based on their proposed solution.

Solutions may need to consider relationships between entities such as:

- Students or student cohorts
- Faculty members
- Modules
- Rooms and academic facilities
- Timetable sessions
- Examination sessions
- Time slots

The exact database structure, relationships, and system architecture are left to the participants.

7. Optional Advanced Challenge Areas

Teams seeking to extend their solution may explore advanced automation or optimization capabilities such as:

- Automated generation of clash-free timetable options
- Constraint-based scheduling
- Examination venue optimization
- Invigilator allocation
- Resource utilization optimization
- Scheduling bottleneck prediction

These capabilities are optional and should not be considered mandatory requirements for a complete submission.

8. Administrative Capability

Administrative functionality may be included where relevant to the proposed solution.

Possible examples include:

- Managing rooms and academic resources
- Configuring scheduling rules or constraints
- Manually resolving or overriding scheduling conflicts
- Adjusting faculty allocations
- Maintaining records of important scheduling changes

Teams are not expected to implement a complete production-level administrative system within the hackathon.

9. Expected Deliverables

Teams should submit:

- **Working Prototype:** A functional prototype demonstrating the proposed solution to the academic scheduling or resource allocation problem.
- **Core Workflow:** At least one meaningful end-to-end scheduling, allocation, optimization, or conflict-management workflow.
- **Solution Demonstration:** Evidence showing how the proposed solution improves the existing manual process.
- **Technical Overview:** A brief explanation of the system architecture, major technologies, scheduling logic, algorithms, or AI techniques used.

Teams are not required to implement every scheduling domain described in the problem statement.

Depth, effectiveness, innovation, and the quality of the demonstrated solution are more important than the number of features implemented.

FINAL DESIGN PRINCIPLE

Do not build only a passive system that stores academic information.

The solution should actively improve at least one aspect of academic scheduling or resource allocation through automation, conflict prevention, optimization, improved visibility, or better decision-making.

The suggested features and solution areas are provided as context rather than as a prescribed system design. Participants are free to determine their own architecture, features, workflows, and technical approach.

Submissions will be judged primarily on how effectively they address the identified problem, the quality of the implemented prototype, technical reasoning, innovation, and practical applicability.